

SKILLZENO INTERNSHIP

TASK 1: EV SYSTEM STUDY

Technical Report on Major Electric Vehicle Components and System Interaction

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1. Introduction

Electric Vehicles (EVs) use electrical energy stored in a battery pack to drive one or more electric motors. Unlike conventional internal-combustion vehicles, EVs do not require a fuel tank, engine, exhaust system, or multi-stage mechanical powertrain. The core EV architecture is built around the traction battery, power electronics, electric motor, control unit, charging system, and supporting electrical and mechanical subsystems.

The objective of this study is to identify the major EV components, explain the role of the battery, motor, inverter, controller, and charger, and show how these systems interact during charging, acceleration, cruising, and regenerative braking.

2. Major Components of an Electric Vehicle

- **Traction Battery Pack:** Stores high-voltage DC electrical energy and supplies it to the traction system. Modern EVs commonly use lithium-ion battery cells grouped into modules and packs.
- **Battery Management System (BMS):** Monitors cell voltage, temperature, current, state of charge, and safety limits. It balances cells and protects the battery from over-charge, over-discharge, overheating, and excessive current.
- **Electric Traction Motor:** Converts electrical energy into rotational mechanical energy to drive the wheels. It can also operate as a generator during regenerative braking.
- **Inverter / Power Electronics:** Converts DC from the battery into controlled AC for the motor and, during regenerative braking, converts generated AC back to DC for the battery.
- **Vehicle Control Unit (VCU):** Acts as the supervisory controller. It processes driver commands and sensor information, then coordinates torque demand, motor operation, battery limits, braking, and safety interlocks.
- **On-board Charger (OBC):** Converts external AC supply to regulated DC suitable for charging the traction battery. In DC fast charging, much of the conversion occurs in the external charging station.
- **DC-DC Converter:** Steps high-voltage battery power down to low-voltage levels, typically for the 12 V or 48 V auxiliary electrical system.
- **Thermal Management System:** Maintains the battery, motor, and power electronics within safe and efficient temperature ranges using air or liquid cooling/heating.
- **Reduction Gear / Differential:** Transfers motor torque to the wheels at a suitable speed and torque ratio. EVs generally need fewer gear ratios because electric motors operate efficiently over a wide speed range.

3. Roles of the Key EV Components

3.1 Battery

The traction battery is the main energy source of the EV. It stores energy in DC form and determines much of the vehicle range, power capability, charging time, weight, and cost. The BMS continuously checks the condition of the battery and communicates allowable charging and discharging limits to the vehicle controller.

3.2 Electric Motor

The motor produces torque directly from electrical current. Common EV motor technologies include permanent-magnet synchronous motors, induction motors, and switched-reluctance motors. Electric motors provide high starting torque, smooth speed control, and high

efficiency. During regenerative braking, the motor reverses its energy flow and acts as a generator.

3.3 Inverter

The battery provides DC power while most high-performance traction motors require controlled three-phase AC. The inverter uses semiconductor switching devices to vary voltage and frequency, thereby controlling motor speed and torque. It also supports reverse power flow during regeneration.

3.4 Controller / VCU

The controller is the decision-making layer of the powertrain. It receives signals from the accelerator pedal, brake pedal, wheel-speed sensors, battery system, thermal system, and safety circuits. It then sends torque commands and operating limits to the inverter, motor, and other ECUs.

3.5 Charger

The charging system transfers energy from the electrical grid to the EV battery. An on-board charger handles AC charging by rectifying and regulating the input. DC fast charging supplies regulated DC directly to the battery through communication and protection systems, enabling much higher charging power.

4. EV System Block Diagram

The simplified block diagram below shows the main energy and control paths in a battery electric vehicle. Solid arrows represent the main power or command flow, while the dashed path indicates regenerative energy recovery.

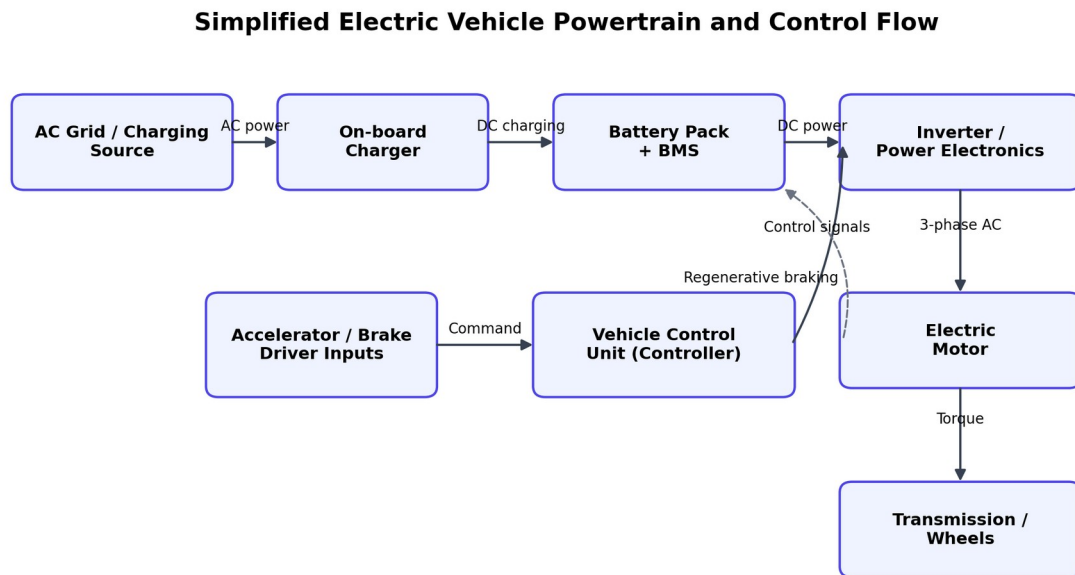


Figure 1: Simplified EV powertrain and control interaction.

5. How the EV Systems Interact

1. **Charging:** The charging source supplies electrical energy. During AC charging, the on-board charger converts AC into DC, and the BMS supervises safe charging of the battery. During DC fast charging, high-power DC is supplied directly to the battery through controlled contactors and communication protocols.
2. **Acceleration:** When the driver presses the accelerator, the VCU calculates the requested wheel torque. The inverter draws DC energy from the battery and produces the required three-phase AC for the motor. The motor generates torque, which is transmitted through the reduction gear and differential to the wheels.
3. **Cruising:** At steady speed, the controller continuously adjusts motor torque to overcome rolling resistance, aerodynamic drag, and road gradient while attempting to keep the system within efficient operating regions.
4. **Regenerative Braking:** When the driver releases the accelerator or applies braking, the controller can command negative motor torque. The motor then acts as a generator, converting vehicle kinetic energy into electrical energy. The inverter routes this energy back to the battery, subject to battery charge limits and traction conditions.
5. **Thermal and Safety Coordination:** Battery temperature, motor temperature, inverter temperature, insulation status, and current limits are continuously monitored. If a limit is approached, the controller can reduce power, limit charging, or disconnect the high-voltage system to protect the vehicle and occupants.

6. Summary Table

Component	Main Function	Energy / Signal Type
Battery Pack	Stores and supplies traction	High-voltage DC

	energy	
Motor	Converts electrical energy to mechanical torque	AC electrical -> mechanical
Inverter	Controls power conversion for the motor	DC <-> controlled AC
Controller / VCU	Coordinates torque, safety, and operating limits	Digital control signals
On-board Charger	Converts grid AC to battery charging DC	AC -> DC
BMS	Monitors and protects battery cells	Sensor data / control signals

7. Advantages of the EV Powertrain

- High energy-conversion efficiency compared with conventional engine-based powertrains.
- Instant and precisely controllable motor torque.
- Regenerative braking can recover part of the vehicle kinetic energy.
- Fewer moving mechanical parts can reduce routine maintenance requirements.
- Zero tailpipe emissions for battery electric vehicles.
- Power electronics and software allow advanced traction, safety, and energy-management functions.

8. Conclusion

An electric vehicle is an integrated electro-mechanical system in which the battery stores energy, the inverter controls energy conversion, the electric motor produces traction, the controller coordinates vehicle behavior, and the charger replenishes the battery from an external power source. Reliable communication among the BMS, VCU, inverter, charger, and thermal systems is essential for efficiency, performance, and safety. Understanding these interactions provides the foundation for studying advanced EV topics such as battery sizing, motor control, regenerative braking, thermal management, charging infrastructure, and vehicle energy management.

9. Learning Outcomes

- Identify the major subsystems of a battery electric vehicle.
- Explain the function of the battery, motor, inverter, controller, and charger.
- Describe energy flow during charging, propulsion, and regenerative braking.
- Interpret a basic EV system block diagram and the interaction between power and control systems.